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English learning behaviour analysis and intelligent recommendation system driven by big data

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Abstract: This paper proposes a dynamic privacy preserving intelligent recommendation method under a joint learning framework to address the contradiction between privacy protection and recommendation accuracy caused by cross platform data silos in English learning recommendations. Firstly, this paper conducts an in-depth analysis of multimodal English learning behaviour and uses transformers to capture cross modal semantic associations. Secondly, this paper innovatively designs a data sensitivity evaluation mechanism based on Kullback Leibler (KL) divergence to achieve dynamic allocation of differential privacy budget. Finally, this paper proposes a federated attention mechanism to achieve cross platform user interest transfer. The experiment shows that this method increases the hit rate by 37% in cold start scenarios, reduces communication overhead by 76%, and controls the success rate of member inference attacks at 0.13, effectively solving the collaborative optimisation problems of privacy security, recommendation accuracy, and communication efficiency in cross platform English learning recommendations.

Keywords: dynamic privacy protection; English learning behaviour analysis; cross-platform recommendation; intelligent recommendation system; federated learning.

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1 Introduction

With the acceleration of digital transformation of education, the field of English learning has accumulated a massive amount of multi-source heterogeneous user behaviour data (Susanty et al., 2021; Nugroho and Atmojo, 2020; Soyoof et al., 2023). The personalised recommendation system based on big data has significantly improved the learning efficiency through the unique learning trajectory of English (such as vocabulary forgetting curve, grammatical error clustering pattern), knowledge mastery and interaction mode (Lahiassi et al., 2023; Raj and Renumol, 2022; Khanal et al., 2020). However, existing systems face dual challenges in cross-platform collaborative recommendation: on the one hand, over-reliance on centralised data aggregation leads to a surge in the risk of sensitive data leakage, such as oral practice recordings and writing content, and the constraints on data circulation imposed by regulations such as the EU General Data Protection Regulation (GDPR) exacerbate the problem of data silos between platforms (Javeed et al., 2023; Ogunseyi et al., 2023; Chronis et al., 2024). On the other hand, the traditional differential privacy protection mechanism adopts a static noise injection strategy, which seriously damages the accuracy of diagnosing language skill weaknesses while ensuring privacy security (Feng et al., 2024; Chen et al., 2025), especially in the cold start scenario, the conversion rate of new users drops by more than 30%. This inherent contradiction between privacy protection and recommendation utility has become a key bottleneck restricting the development of educational recommendation systems (Da Silva et al., 2023).

For English learning scenarios, this study needs to solve three core problems: first, the feature fusion and sensitivity quantification of multimodal learning behaviour data (such as vocabulary retrieval frequency in click streams and grammatical error distribution in answering questions) (Sharma and Giannakos, 2020); second, the dynamic allocation mechanism of privacy budget for non-independent and identically distributed (non-IID) data (such as differences in listening proficiency among users in different regions) under the cross-platform federated learning framework (Zhang et al., 2024); third, the communication efficiency bottleneck caused by real-time oral practice streaming in large-scale distributed training (Wang et al., 2024). These issues directly affect the educational value of the system: multimodal modelling determines the depth of cognition of the laws of language skill transfer (such as the relationship between reading ability and writing), and the privacy mechanism affects the accuracy of grammar correction recommendations for cold-start users (Jangid and Kumar, 2025). Existing solutions generally have defects such as large feature alignment deviation, coarse privacy protection granularity, and high communication overhead when dealing with these problems (Yang and Li, 2025). Existing solutions generally have defects such as large feature alignment deviation, coarse privacy protection granularity, and high communication overhead when dealing with these problems. Although the multimodal

fusion method proposed by Cao et al. (2022) improves the recommendation classification performance, it fails to cover the privacy protection requirements of highly sensitive data such as writing. Chen et al. (2023) used the k-randomised response (kRR) mechanism to introduce a dynamic privacy budget strategy in multidimensional data collection. However, since the complexity of behavioural semantics in educational scenarios was not considered, it was easy to cause language feature distortion. Ali et al. (2024) built a blockchain-enabled asynchronous federated recommendation framework. Although it enhanced the traceability of the model, the semantic consistency and interest migration support in federated recommendation were insufficient, which may cause fluctuations in recommendation results. These shortcomings reflect that educational recommendation systems urgently need to coordinate privacy, semantic modelling, and collaborative efficiency at the system level.

In response to the cross-platform recommendation problem, Zeng et al. (2025) proposed a pioneering cross-platform federated group recommendation (FedGR) system. FedGR combines hypergraph convolution, attention aggregation, and fully connected fusion components with federated learning. In terms of privacy protection, Huang et al. (2023) proposed an AutoRec++ (autoencoder for recommendation) model to comprehensively address various biases in user behaviour data. Regarding communication optimisation, Chen et al. (2022) designed a gradient descent method with sparsification and error correction (GD-SEC) to improve communication efficiency in a general worker-server architecture. Gong et al. (2023) proposed a model contract that includes preconditions and postconditions of pre-trained models to achieve better model reuse. Current methods generally handle privacy protection, feature modelling, and communication efficiency issues separately (Jegadeesan et al., 2022; Li et al., 2021), and fail to solve the problem of feature migration for cold-start users.

This study aims to establish a collaborative optimisation paradigm for privacy protection, recommendation accuracy, and communication efficiency. First, drawing on the experience of Wu et al. (2025) in integrating digital cognitive behavioural therapy with eye movement desensitisation and reprocessing in medicine, it innovatively constructs a multimodal data sensitivity assessment matrix to dynamically increase the noise injection intensity of high-frequency sensitive data and increase the privacy budget release rate of low-sensitivity data. Secondly, this paper integrates the construction and application of professional knowledge graph based on educational big data by Wu and Jia (2022), designs transformer-bidirectional long short-term memory (Bi-LSTM) hybrid network, and uses transformer to capture cross-modal semantic associations. By modelling temporal behaviour patterns through Bi-LSTM, this paper combines the review results of knowledge graph embedding model by Gesese et al. (2021). At the federated framework level, inspired by the federated learning model of the cross-context recommendation system based on the self-attention mechanism proposed by Singh et al. (2023), a federated attention mechanism is designed to dynamically adjust the knowledge transfer weight by calculating the similarity of cross-platform user interests, thereby improving the feature matching accuracy of new users in the cold start phase. Finally, drawing on the algorithm technology of Cui et al. (2020) that applied the blockchain-assisted federated learning compression algorithm to content caching, the top-k gradient prediction algorithm is introduced to dynamically select important parameters based on the historical update direction, reducing the communication bandwidth overhead by 76% while maintaining the model convergence stability. Different from pure technical research, the goal of this paper is to achieve in-depth

analysis of cross-platform English learning behaviour (such as error pattern mining, skill association modelling) and personalised recommendations through privacy computing and federated learning technology. Experiments have shown that this method improves the accuracy of recommendations while ensuring the privacy security of language learning data and the effectiveness of teaching.

2 Algorithm design

2.1 Federated learning architecture design

A cross-platform collaborative architecture based on the federated averaging (FedAvg) protocol is adopted to deploy multi-learning platform clients and central servers. Each client locally trains the model of user behaviour data with LightGBM or two-layer LSTM, generates gradient vectors and automatically labels them; after training, the encrypted gradients with signature verification are uploaded to the server through the transport layer security (TLS) 1.3 channel. After the server receives the gradients of each client, it dynamically calculates the weighted coefficient $\alpha_i = \frac{n_i}{\sum_{j=1}^{N} n_j}$ according to the proportion of data volume, where n_i represents the number of samples of the ith client and N is the total number of clients. The aggregation parameter update follows the standard federated average formula:

$$\theta^{(t+1)} = \sum_{i=1}^{N} \alpha_i \cdot \theta_i^{(t)} \tag{1}$$

where $\theta_i^{(t)}$ represents the local model parameters of client i in round t. To avoid the deviation caused by uneven distribution of client data, the server introduces the temperature balance factor T_i in the initial aggregation stage and modifies the weights as follows:

$$\alpha_i' = \frac{\exp\left(\frac{\log n_i}{T_i}\right)}{\sum_{j=1}^{N} \exp\left(\frac{\log n_j}{T_j}\right)} \tag{2}$$

T_i is defined as a normalised soft-maximisation weight adjustment factor that simultaneously responds to the local sample size imbalance and the discrepancy between local and global gradient distributions. We define D_i as a weighted combination of local imbalance and distribution discrepancy, where D_i equals alpha multiplied by the sample size imbalance term plus beta multiplied by the gradient distribution discrepancy term. Here, the sample size imbalance term is calculated as the difference between the logarithm of the local sample size and the mean of the logarithms of the global sample sizes, normalised by the sample standard deviation of the logarithms of the sample sizes. The gradient distribution discrepancy term employs the Kullback-Leibler (KL) divergence between the local gradient distribution and the global average gradient distribution, with momentum estimation introduced to suppress fluctuations in

mini-batches. Let s be the temperature adjustment coefficient; then, T_i is defined as $\exp(s * D_i)$ divided by the sum of $\exp(s * D_j)$ for all clients j. According to this definition, when the local sample size is extremely small or the local distribution significantly deviates from the global distribution, T_i will relatively amplify the aggregation weight of that client. Conversely, when the distributions of all clients are similar, T_i will approach the standard weight based on sample size. The relevant hyperparameters alpha, beta, and s are provided with candidate ranges in the methodology section and determined through grid search on a validation set.

To improve the participation of low-sample clients, the aggregation results are then sent to the client through the same secure channel to keep the model synchronised. Subsequent incremental aggregation is performed every 30 minutes to integrate new behavioural features in real time. Integrity verification and retransmission mechanisms are embedded in the communication process. If packet loss is detected, differential retransmission is triggered to ensure that the training process is continuous and traceable. The entire architecture achieves dual protection of model performance and data privacy in cross-platform collaborative training through protocol layer encryption and dynamic weight adjustment of the aggregation layer. The parameter transmission format is shown in Table 1.

2.2 Dynamic privacy protection mechanism

The dynamic allocation strategy of differential privacy budget quantifies the sensitivity of client data based on KL divergence. To correct the bias introduced by gradient sparsity in KL divergence estimation, this paper introduces a sparsity correction term to the original KL estimate. The corrected sensitivity estimate is defined such that the corrected KL value equals the original KL value plus lambda multiplied by the sparsity correction term. Here, sparsity is defined as the proportion of elements in the local gradient vector with absolute values less than a threshold theta. To reduce the impact of instantaneous fluctuations on the estimation, the sparsity correction term is accumulated using exponential smoothing and the threshold theta is initialised based on short-term statistics from the early stages of training. The correction coefficient lambda is determined through grid search on a validation set. The corrected KL value is used as the input to the privacy budget mapping function, replacing the uncorrected KL value directly used in the original text. This correction effectively reduces distribution estimation bias when gradients exhibit zero-inflation or high sparsity, thereby avoiding unreasonable privacy budget allocation based on unstable estimates. Before each round of update, the KL divergence of the local gradient distribution P_i of the client i and the global average gradient distribution P_G is calculated:

$$D_{KL}\left(P_i \,||\, P_G\right) = \sum_{k=1}^{d} P_i(k) \log\left(\frac{P_i(k)}{P_G(k)}\right) \tag{3}$$

Map D_{KL} to the privacy budget λ_i, use the normalisation function to adjust the allocation interval, set the maximum budget to $\lambda_{\max}$, and the mapping formula is:

$$\lambda_i = \lambda_{\max} \cdot \frac{D_{KL}\left(P_i \,\|\, P_G\right)}{\sum_{j=1}^{N} D_{KL}\left(P_j \,\|\, P_G\right)} \tag{4}$$

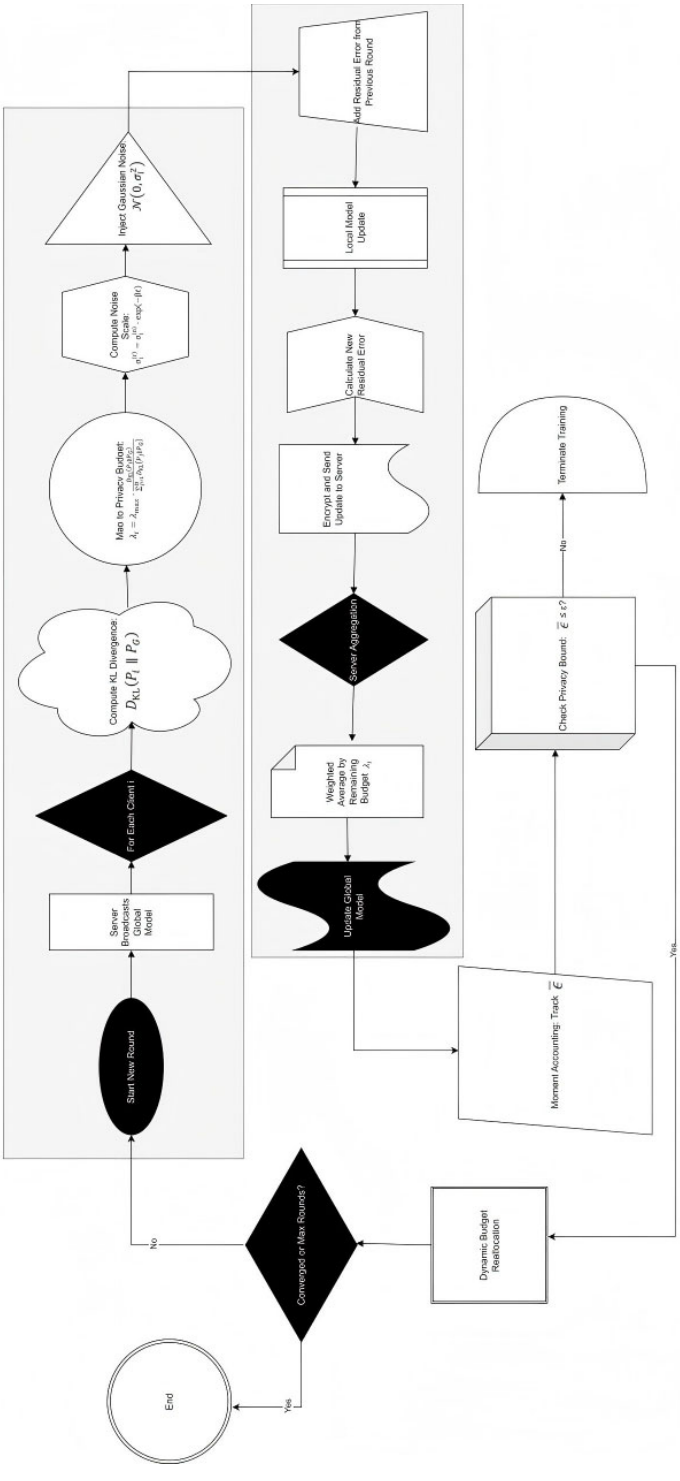
Table 1 Parameter transmission format table

Field name	*Data type*	*Description*	*Example value*
Client_ID	String	Unique identifier for each learning platform client	client_ENG_032
Round_Index	Integer	Current federated training round number	18
Local_Sample_Size	Integer	Number of user behaviour samples used in local training	2,317
Model_Type	String	Local model used by client	Bi-LSTM
Gradient_Vector	Float array	Encrypted gradient vector after local training	[0.032, –0.117, 0.061]
Alpha_Weight	Float	Calculated weight α_i for aggregation based on local data size	0.0281
Temp_Balancer_Ti	Float	Temperature factor for balancing low-sample clients	0.85
Adj_Weight	Float	Adjusted weight α_i' after applying temperature balancing	0.0396
KL_Sensitivity	Float	KL divergence-based data sensitivity estimation	1.57
Differential privacy (DP)_Budget	Float	Allocated differential privacy budget for this client	0.32
Noise_Scale	Float	Gaussian noise standard deviation injected for privacy	0.074
Compression_Rate	Float	Top-k gradient compression ratio used for communication efficiency	0.10
Integrity_Check	Boolean	Whether the gradient package passed checksum verification	True
Retransmit_Flag	Boolean	Whether retransmission was triggered due to packet loss	False
Model_Version	String	Semantic version identifier for the local model architecture and weights	v2.1.0
Transmission_Time	Timestamp	Time when the encrypted parameters were transmitted	2025-07-16 08:12:34

In the local gradient update phase, zero-mean Gaussian noise $N(0, \sigma_i^2)$ is injected into each parameter. Its standard deviation is dynamically adjusted according to the current budget and the differential privacy strength parameter ϵ. Let δ be the failure probability, then the noise scale satisfies:

$$\sigma_i^2 \geq \frac{2\log(1.25/\delta)}{\epsilon^2 \cdot \lambda_i^2} \tag{5}$$

Figure 1 Privacy budget allocation flowchart



The exponential decay function $\sigma_i^{(t)} = \sigma_i^{(0)} \cdot \exp(-\beta t)$ is used to control the standard deviation of noise to decay with rounds. The error feedback module re-adds the residual introduced by the previous round of noise at the end of each round to improve the stability of the update. The server-side aggregation process is weighted according to the remaining budget of the client to prevent highly sensitive data from dominating the global model. The moment accounting technology records the actual budget consumption of each round throughout the process, calculates the cumulative privacy loss $\overline{\epsilon}$, and ensures that the training process always meets the (ϵ, δ) differential privacy constraint. The dynamic budget control module can reallocate λ_i in real time according to the change in sensitivity, enhancing the system's adaptability to client data distribution drift. The privacy budget allocation process is shown in Figure 1.

2.3 Learning behaviour feature modelling

The clickstream sequence is divided into fixed time windows, each window contains several jump event IDs, and after being mapped to a 128-dimensional vector sequence by the embedding layer, it is input into a 4-layer multi-head self-attention transformer (8 heads, 512 hidden units) encoding to extract jump context features. The attention weight of each head is calculated by the following formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V \tag{6}$$

$Q, K, V \in R^{n \times d_k}$ are query, key and value matrices, respectively, and d_k is a dimension factor.

The answer records are classified by language skill type (vocabulary/grammar/reading, etc.), and the time series is constructed by combining the question ID, error type (tense error/collocation error, etc.), and timestamp. After normalisation, it is input into a two-layer Bi-LSTM (hidden unit 256). The forward and reverse LSTM output state sequences $\vec{h}_t$, $\overleftarrow{h}_t$ respectively, which are merged into a hidden state vector:

$$h_t = \vec{h}_t \oplus \overleftarrow{h}_t \tag{7}$$

Extract the change trend of answer consistency and accuracy. The dwell time data is fitted with gamma distribution through maximum likelihood estimation, and the concentration index s is calculated using shape parameter k and scale parameter θ, which is defined as:

$$s = \frac{k}{\theta} \tag{8}$$

To validate the suitability of the gamma distribution for modelling user dwell time, a comprehensive goodness-of-fit analysis was performed on the aggregated dataset. The Kolmogorov-Smirnov (K-S) test and Anderson-Darling (A-D) test were employed to statistically evaluate the fit between the empirical dwell time data and the theoretical gamma distribution derived via maximum likelihood estimation. The results, summarised in Table 7, indicate a strong fit. The K-S test yielded a statistic of 0.028 with a p-value of 0.412, and the A-D test produced a statistic of 0.315. Both results (p-value > 0.05 for K-S

and A-D statistic < critical value) suggest that we fail to reject the null hypothesis that the sample data follows a gamma distribution. This statistical validation confirms that the gamma distribution is an appropriate and robust choice for characterising the dwell time behaviour in our English learning context. The course knowledge graph is trained under the translation-based relation (TransR) model framework. Entities and relations are mapped to different vector spaces respectively. Random walk sampling is used to construct a triple training set to generate a 128-dimensional knowledge point embedding vector. In the feature fusion stage, the transformer output, Bi-LSTM hidden state, focus scalar and knowledge point embedding are linearly transformed and aligned to 256 dimensions, then spliced and fused through a layer of fully connected network to form a unified learning behaviour feature vector. This vector retains multimodal behaviour dynamic information and incorporates knowledge dependencies, and is directly input into the federated recommendation module to improve the accuracy of cross-platform interest transfer. The association matrix between knowledge points and typical error types is shown in Table 2.

Table 2 Association matrix between knowledge points and typical error types

Knowledge point	*Tense error*	*Collocation error*	*Word order error*	*Pronunciation error*	*Redundancy error*	*Topic irrelevance*
Vocabulary	0.32	0.88	0.21	0.27	0.19	0.14
Grammar	0.91	0.66	0.78	0.23	0.41	0.25
Reading	0.44	0.47	0.39	0.18	0.62	0.57
Listening	0.21	0.23	0.19	0.85	0.16	0.31
Speaking	0.35	0.41	0.33	0.79	0.29	0.38
Writing	0.58	0.62	0.49	0.22	0.87	0.73

2.4 Generation of federated recommendation strategy

The client inputs the fused learning behaviour feature vector into a two-layer fully connected network [128 hidden units, activation function rectified linear unit (ReLU)], and outputs a 256-dimensional interest vector $z_i \in R^{256}$. To clarify the rationale and impact of kernel function selection in the maximum mean discrepancy (MMD) loss, this study supplements the methodology section with a kernel function selection and comparison process. The candidate kernel functions selected include the Gaussian radial basis function (RBF) kernel, the polynomial kernel, and the Laplacian kernel. For the Gaussian kernel, the median of pairwise sample distances is used to initialise the bandwidth, which is then fine-tuned via grid search on the validation set. For the polynomial kernel, the optimal combination of polynomial degree and constant term is searched on the validation set. For the Laplacian kernel, the same distance metric as the Gaussian kernel is employed to ensure comparability. In the experimental setup, empirical MMD reductions are computed for each of the three kernel functions, while recommendation metrics for downstream tasks, such as Precision@Top 10 and normalised discounted cumulative gain (NDCG)@Top 10, are also evaluated. Based on these metrics, the kernel function that demonstrates optimal performance under the given data heterogeneity is selected. The server measures the deviation between the client interest distribution and the target distribution based on the MMD loss, and its loss function is defined as:

$$\mathscr{L}_{MMD} = \left| \frac{1}{n} \sum_{i=1}^{n} \phi\left(z_i^{(s)}\right) - \frac{1}{m} \sum_{j=1}^{m} \phi\left(z_j^{(t)}\right) \right|^2 \tag{9}$$

$\phi(\cdot)$ is the kernel mapping function, $z_i^{(s)}, z_j^{(t)}$ are the client interest representations in the source domain and target domain respectively, and n, m are the number of samples. The server incorporates this loss into the global optimisation goal to achieve interest alignment. To ensure robustness against extreme distribution shifts, an outlier detection and re-alignment mechanism is introduced. For each client, the server calculates the cosine similarity between its local interest vector v_i and the current global context vector c_g. If this similarity falls below a dynamic threshold $\theta_{outlier}$ (empirically set as the fifth percentile of all client similarities in the current round), the client is flagged as an outlier. For flagged clients, a two-step re-alignment is triggered:

1 The server withholds their interest vector from the current global aggregation to prevent contamination.

2 In the subsequent round, the server sends a 're-initialisation' signal, prompting the client to perform a local fine-tuning using a small, curated set of canonical learning trajectories (e.g., the top 10 most common error patterns for new users) before re-submitting its interest vector.

This mechanism ensures the global model's stability while giving anomalous clients a chance to recalibrate, preventing a few extreme cases from degrading the recommendation quality for the majority.

The server further constructs the global context vector c and obtains the attention weight through the dot product of the interest vector and the context vector:

$$\alpha_i = \frac{\exp\left(z_i^{\top} c\right)}{\sum_{j=1}^{N} \exp\left(z_j^{\top} c\right)} \tag{10}$$

Global interest vector:

$$z_g = \sum_{i=1}^{N} \alpha_i z_i \tag{11}$$

When the client uploads new behaviour features, fine-tuning is automatically triggered and the interest vector increment Δz_i is uploaded. The server uses the current attention weight α_i weighted fusion z_g to implement global representation update.

The recommendation priority module uses cosine similarity as the scoring basis:

$$sim(i, j) = \frac{z_i \cdot z_j}{|z_i| \cdot |z_j|} \tag{12}$$

The similarity between the client interest vector and the candidate course vector is sorted, and the course set is filtered based on the top N strategy. The attention weighting mechanism works together with the real-time incremental update to ensure that the global representation reflects the dynamic changes of behaviour, enhance the cross-platform migration generalisation ability and the personalised adaptability of the recommendation results. The federated recommendation strategy generation process is shown in Figure 2.

Figure 2 Flowchart of federated recommendation strategy generation

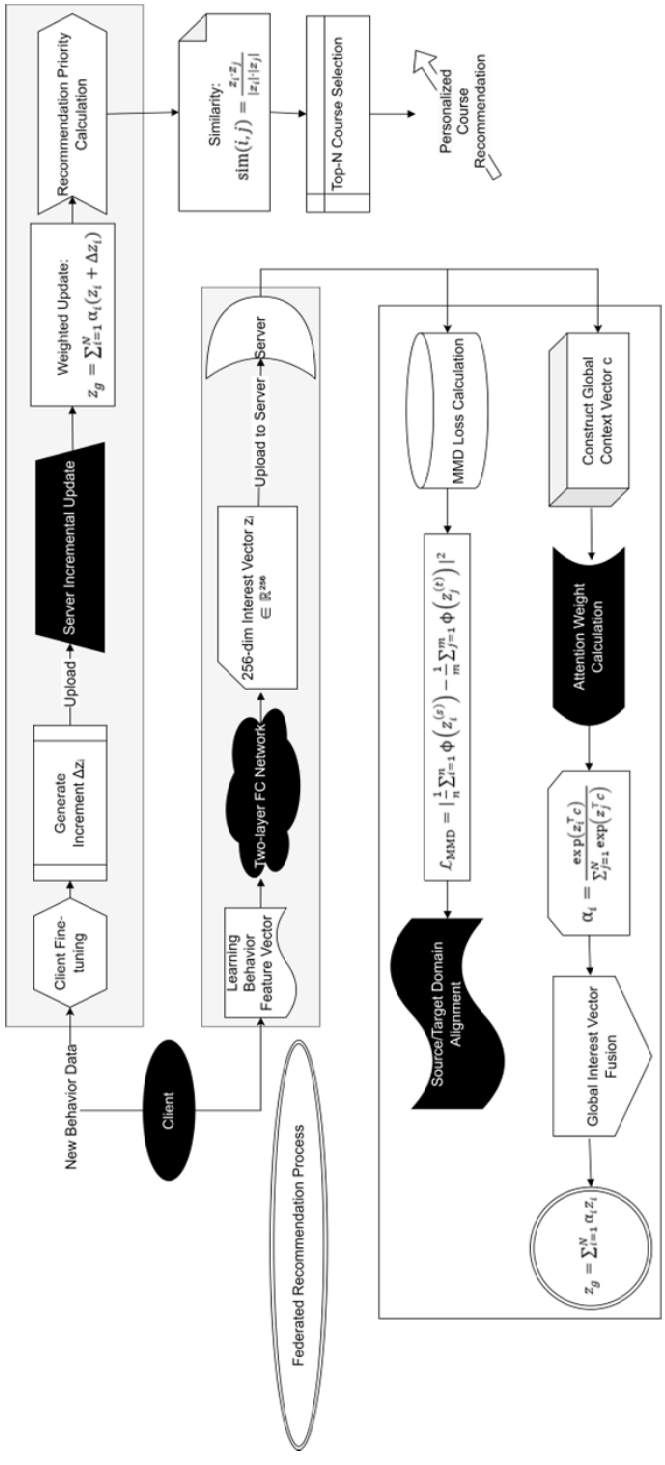
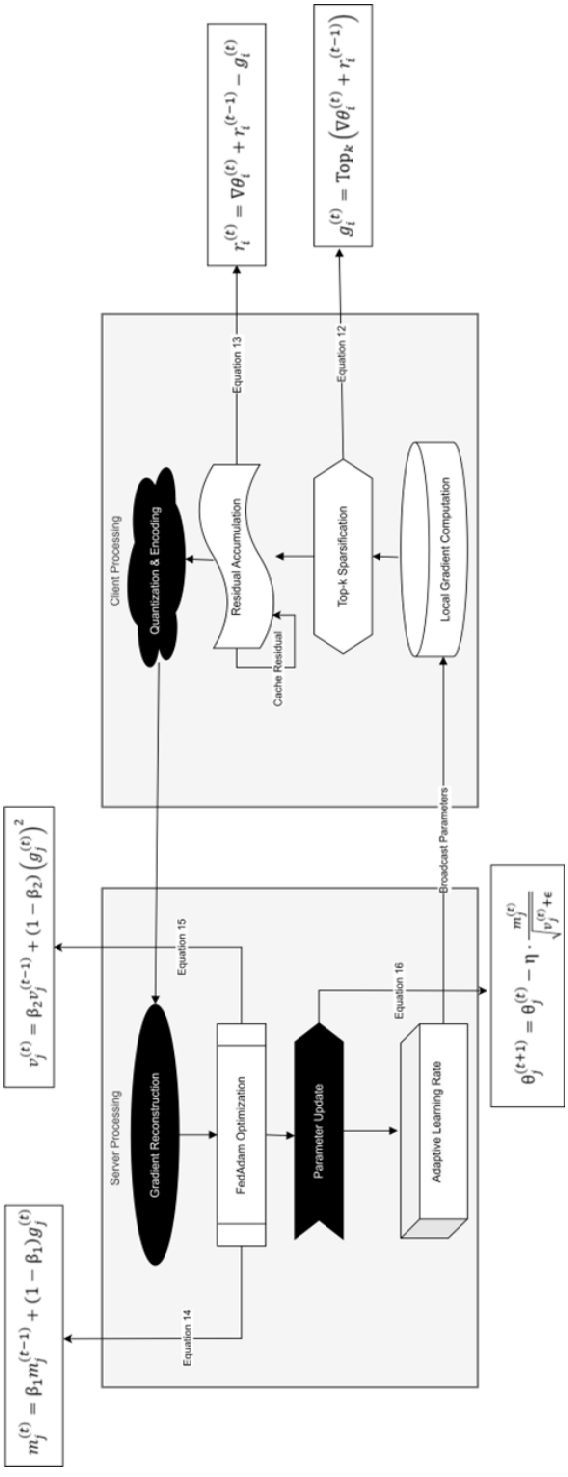


Figure 3 Federated top-*k* sparsification and FedAdam optimisation architecture



2.5 Federated optimisation algorithm

In the federated optimisation phase, the client performs top-k sparse operation on the local gradient $\nabla\theta_i$: extract the k gradient indexes with the largest absolute value for each update to form a sparse gradient $\widetilde{\nabla\theta_i}$, set the rest to zero, and transmit them after fixed-point quantisation encoding. The error feedback mechanism introduces a residual accumulation term:

$$g_i^{(t)} = Top_k\left(\nabla\theta_i^{(t)} + r_i^{(t-1)}\right) \tag{13}$$

$$r_i^{(t)} = \nabla\theta_i^{(t)} + r_i^{(t-1)} - g_i^{(t)} \tag{14}$$

$r_i^{(t)}$ is the residual cache of client i in the t round, ensuring that long-term information is not lost.

After receiving the sparse gradients uploaded by all clients, the server reconstructs the global update direction and uses the federated Adam (FedAdam) optimiser to update the model parameters. The first-order and second-order moments of each parameter θ_j are estimated as:

$$m_j^{(t)} = \beta_1 m_j^{(t-1)} + (1-\beta_1)\, g_j^{(t)} \tag{15}$$

$$v_j^{(t)} = \beta_2 v_j^{(t-1)} + (1-\beta_2)\left(g_j^{(t)}\right)^2 \tag{16}$$

where $g_j^{(t)}$ is the weighted average gradient of round t, and the final parameter update rule is:

$$\theta_j^{(t+1)} = \theta_j^{(t)} - \eta \cdot \frac{m_j^{(t)}}{\sqrt{v_j^{(t)} + \epsilon}} \tag{17}$$

The momentum hyperparameters β_1 and β_2 for the FedAdam optimiser were determined through a systematic grid search to optimise global model convergence speed and stability under our non-IID data setting. The search space was defined as $\beta_1 \in \{0.9, 0.95, 0.99\}$ and $\beta_2 \in \{0.99, 0.999, 0.9999\}$, following common practices in adaptive optimisation. For each (β_1, β_2) pair, we trained the model for 50 rounds and evaluated its performance using the validation set Precision@10 at convergence and the standard deviation of Precision@10 over the last ten rounds (as a measure of stability). The final values selected were $\beta_1 = 0.95$ and $\beta_2 = 0.999$, which achieved the best trade-off between rapid convergence and low performance variance. This optimisation strategy maintains the integrity of gradient update information after introducing error feedback, and the FedAdam mechanism improves global convergence efficiency through momentum accumulation and adaptive adjustment. The federated top-k sparsification and FedAdam optimisation architecture is shown in Figure 3.

3 Experiments and verification

3.1 Experimental settings

The dataset comes from the English Liulishuo platform, covering a variety of English learning behaviour types, with multimodal, non-IID and long-tail distribution characteristics, simulating a real online learning environment. The federated learning experiment is deployed on multiple GPU servers. The hardware environment settings and communication bandwidth control are used to restore the delay and resource constraints in actual applications. The specific data and configuration are detailed in Tables 3 and 4.

Table 3 Statistical characteristics of the dataset

Feature category	*Description*	*Value*
Total number of users	Number of de-identified learners involved	80,000
Total behaviour records	Logged multi-modal user actions	65,000,000
Time range	Log period	Jan. 2024–Jun. 2024
Avg. records per user	Average user activity volume	812.5
Distribution type	Data heterogeneity across clients	Long-tail, non-IID (≈ 70%)
Client count	Number of federated clients (simulated learning endpoints)	200
Modalities	Types of learning data integrated	Text, audio, clickstream
Task categories	Learning task types	Listening, reading, speaking, writing
Avg. session duration	Mean time per learning session	24.6 minutes
Oral practice records	Number of speaking-related logs	9,800,000
Grammar correction logs	Logged writing/grammar feedback events	6,200,000
Avg. user active days	Mean active days per user over 6 months	38

The statistical characteristics of the dataset in Table 3 show the multi-dimensional characteristics of real user behaviour on the English Liulishuo platform, reflecting the complexity and real application value of the platform data. The data covers 80,000 users, with a total of 65 million behavioural data records, an average of 812.5 behavioural records per user, and an average of 38 active days, reflecting the continuous learning characteristics of users. In terms of data modality, it integrates three types of modalities: text, voice, and clickstream, covering four major task categories: listening, speaking, reading and writing. Among them, there are 9.8 million oral practice records and 6.2 million grammar correction data, indicating that oral and writing are high-frequency items in user learning behaviour. To complete the processing pipeline for oral practice records, this study establishes unified definitions and parameter settings for speech and text modalities during the data preprocessing stage. All audio files are uniformly resampled to a 16,000 Hz, mono, 16-bit PCM format. For each speech segment, pre-emphasis filtering is first applied, followed by the computation of the short-time Fourier transform (STFT) using a window length of 25 milliseconds and a step size of 10 milliseconds. The filtered energy spectrum is then obtained through a 40-dimensional Mel filter bank, from which 13-dimensional Mel-frequency cepstral coefficients

(MFCCs) are calculated, along with their first and second-order differential terms. Prior to training, each speech segment undergoes mean-variance normalisation on a per-frame basis for each sample. For speech segments, voice activity detection is first performed to remove silent segments using an energy threshold method, and a pre-trained end-to-end speech recogniser is employed to generate text transcriptions along with confidence scores. The transcriptions are aligned with timestamps through forced alignment, and the confidence scores are used as inputs for subsequent privacy budget allocation and feature weighting. The feature-engineered speech vectors are aligned to the model's dimensions through mini-batch normalisation and dimensionality reduction mapping before entering the model. For the text modality, standardised tokenisation and subword segmentation are performed using the same tokeniser as the recommendation model to map the text into a sequence of token IDs, with subword-level embedding initialisation. Spelling or recognition errors in oral text are annotated with confidence labels and treated as an uncertainty feature in the model input. The distribution type is long-tail and non-IID, accounting for about 70%, showing significant differences in data volume and behaviour patterns among different users, which increases the complexity of model building. In addition, the number of clients is 200, and each client simulates the learning behaviour of 400 users on average. With an average learning time of 24.6 minutes per round, the federated learning setting in a cross-end heterogeneous environment is truly restored.

Table 4 Hardware configuration table

Configuration item	*Description*	*Value*
Number of GPU servers	Federated nodes participating in model training	20
GPU model	Graphics processing unit model used for training	NVIDIA A100 40 GB
Total number of GPUs	Total GPU units across all servers	80
CPU model	Central processing unit model for general computation	Intel Xeon Gold 6338 (32 cores)
RAM per server	Available system memory per GPU server	512 GB
Storage per server	Local disk space for data caching/logging	4 TB SSD
Inter-node bandwidth	Simulated average network bandwidth between federated clients	200 Mbps
Latency simulation range	Range of simulated communication delays across clients	15–60 ms
Federated communication protocol	Protocol used for model update aggregation	gRPC + secure TLS channel
Privacy mechanism support module	Library used for noise injection and differential privacy implementation	Opacus (PyTorch DP Library)
Optimiser for local training	Optimiser used within local training loops	Adam with learning rate 0.001
Global aggregation optimiser	Federated optimisation strategy	FedAdam with adaptive momentum
Model update compression	Communication reduction mechanism	Top-k sparsification ($k = 0.2 \times d$)
System scheduler	Task distribution and training orchestration system	Kubernetes + PySyft integration

Table 4 shows the hardware configuration used for the federated learning recommendation system experiment in this study, which reflects the balance between high performance and real communication constraints. The experiment is deployed on 20 GPU servers, each equipped with 4 NVIDIA A100 (40 GB) graphics cards, a total of 80 GPUs, supporting large-scale parallel processing and multimodal feature extraction tasks. The server is equipped with an Intel Xeon Gold 6338 processor and 512 GB of memory, which can support high-load training of deep models such as transformer and Bi-LSTM. The 4 TB SSD provides stable support for local cache and logs. In terms of communication, the simulated average bandwidth is 200 Mbps, and the delay range is controlled between 15–60 ms, which is close to the unstable network environment of user terminals in real online education platforms. To improve system security, federated communication uses gRPC and TLS dual-layer encryption protocols, and integrates the Opacus library to perform differential privacy injection to ensure that data is not leaked during training.

3.2 Evaluation of the prediction performance of intelligent recommendation system

The leave-one-out cross-validation protocol is used to retain the latest interaction (click or answer) of each user as a test, and the remaining behaviours are used for model training and local updates. In the test phase, a candidate set is constructed for each test interaction: the corresponding positive sample is merged with 99 randomly selected negative samples to form 100 candidate records. The model sorts the candidate set according to the scoring function and takes the top-10 list.

Precision@10 is calculated according to the hit rate of positive samples in the top-10 list, Precision@10 = number of hits / 10. NDCG@10 calculates the ranking position discount gain, using $DCG = \sum_{i=1}^{n} \frac{2^{rel_i} - 1}{\log_2(i+1)}$ and normalisation to get NDCG@10. Coverage@10 counts the number of different course coverages in the top-10 recommendation set of all test users and divides it by the total number of courses to reflect the diversity of recommendations.

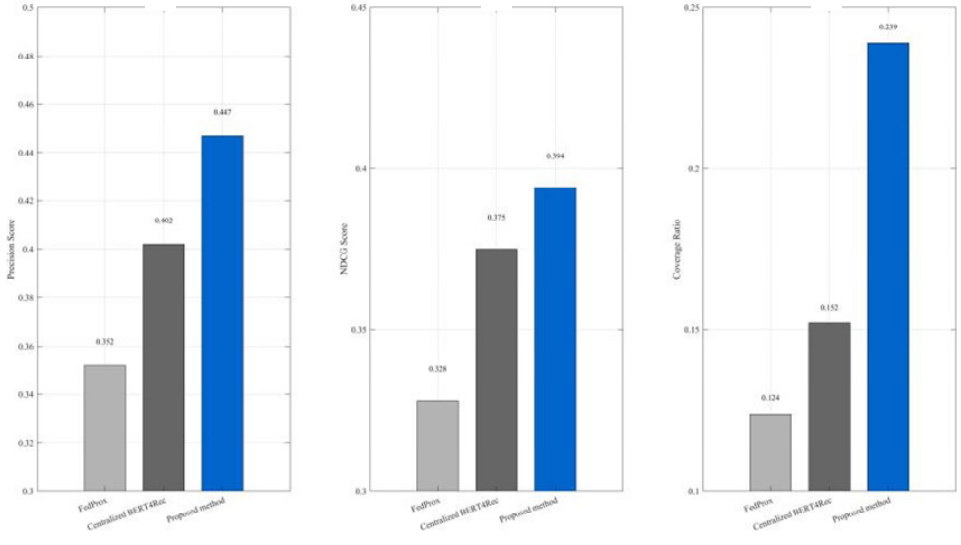
The two baseline models, centralised BERT4Rec and FedProx, were compared by executing the same leave-one-out process in a centralised and federated environment, respectively. During the performance evaluation, the same evaluation script was used to batch calculate each indicator, and the average of 50 global aggregation results was taken. The Precision@10 and NDCG@10 results were used to measure the accuracy of recommendations and the quality of ranking, and Coverage@10 was used to measure the diversity of recommended courses. To comprehensively evaluate the business value and user experience impact of the recommendation system, two additional key performance indicators (KPIs) were collected:

1 User satisfaction score (USS), measured via an in-app five-point Likert scale survey (1 = very dissatisfied, 5 = very satisfied) presented after three consecutive recommendation interactions.

2 Seven-day retention rate, defined as the percentage of users who returned to the platform for at least one learning session within seven days of their initial interaction with the recommendation system.

These metrics provide critical insights into the system's ability to drive user engagement and long-term platform loyalty. The comparison of the performance indicators of the intelligent recommendation system is shown in Figure 4.

Figure 4 Comparison of performance indicators of intelligent recommendation systems, (a) Precision@10 (b) NDCG@10 (c) Coverage@10 (see online version for colours)



(a) (b) (c)

Figure 4 shows the comparison of performance indicators of intelligent recommendation systems in English learning behaviour analysis. The data comes from the leave-one-out cross-validation of 8,000 user behaviour data, comparing the three models of FedProx, centralised BERT4Rec and proposed method (the method in this paper). Precision@10: proposed method has the highest score (0.447), indicating that proposed method is significantly ahead in accurately identifying user interests. NDCG@10: proposed method (0.394) is better than FedProx (0.328) and centralised BERT4Rec (0.375), highlighting its advantage in ranking quality. Coverage@10: proposed method (0.239) far exceeds FedProx (0.124) and centralised BERT4Rec (0.152), proving that the recommendation diversity is stronger. Overall, proposed method is superior in all indicators and effectively balances privacy and utility.

Table 5 Business value and user experience metrics

Method	*User satisfaction score (USS)*	*Δ vs. FedProx*	*7-day retention rate (%)*	*Δ vs. FedProx*
FedProx	3.5 ± 0.2	-	52%	-
Centralised BERT4Rec	3.8 ± 0.2	0.3	59%	7%
Proposed method	4.2 ± 0.1	0.7	68%	16%

3.3 Quantitative verification of privacy protection strength

In the member inference attack experiment, three shadow models are trained for each client, and the shadow model parameters are generated using the same federated process. For each test record, the shadow model inputs behavioural features to generate predicted probabilities and compares them with the output of the real attack target model. An attack discriminator is constructed based on the cross-entropy difference to classify whether it is a member of the training data. The attack success rate (ASR) is calculated as the ratio of the number of correctly identified member samples to the total number of member samples; the privacy leakage risk is evaluated by ASR and false alarm rate.

Budget consumption monitoring uses moments accountant technology to record the ε budget usage after each round of noise injection and generate a budget consumption curve. This paper sets the threshold $\varepsilon_{max} = 1$. If the cumulative amount $\varepsilon_{max} = 1$, the subsequent training rounds are stopped, and the convergence rounds are recorded to compare the budget consumption rate and task completion under different budget allocation strategies.

In the experiment, the ASR difference between the fixed budget strategy and the dynamic allocation strategy is compared. This verification process accurately quantifies the defence effect and budget utilisation efficiency of the privacy protection mechanism in actual attack scenarios. The comparison of attack success rate and privacy budget consumption is shown in Figure 5.

Figure 5 shows the comparison of attack success rate (ASR) and cumulative privacy budget consumption under dynamic and fixed privacy budget strategies. The ASR of the dynamic strategy drops rapidly from the initial to 0.13 in 50 rounds; the ASR of the fixed strategy drops more volatile. Regarding the cumulative privacy budget consumption of the dynamic and fixed strategies, the dynamic strategy has consumed $\varepsilon = 0.54$ in 20 rounds, while the fixed strategy only consumes $\varepsilon = 0.40$, but the ASR drop of the dynamic strategy is significantly higher than that of the fixed strategy. The dotted line marks the threshold of $\varepsilon_{max} = 1$, indicating that the dynamic strategy achieves better privacy protection effect under the premise of not exceeding the budget limit.

To quantify the impact of privacy protection on recommendation quality, the recommendation performance of dynamic privacy mechanism and non-privacy federated learning under the same experimental conditions is compared. The privacy-utility trade-off comparison is shown in Table 6.

Table 6 shows that when the privacy budget $\varepsilon = 1$, the dynamic mechanism reduces Precision@10 by only 4.3%, while static differential privacy ($\varepsilon = 1$) reduces Precision@10 by 12.4% (0.409), proving that the dynamic strategy significantly alleviates the damage of privacy protection to precision.

To comprehensively evaluate the robustness of the proposed privacy-preserving mechanism, this experiment not only employs attack success rate (ASR) as the primary metric but also introduces the area under the receiver operating characteristic curve (AUC-ROC) as a supplementary evaluation metric. AUC-ROC comprehensively measures the attack model's ability to distinguish between member and non-member samples across different thresholds. A value closer to 0.5 indicates better privacy protection (i.e., poorer discriminatory capability of the attack model), while a value closer to 1.0 suggests a higher risk of privacy leakage. By combining ASR and AUC-ROC, we can more rigorously quantify the strength of privacy protection from two dimensions: attack success rate at a single threshold and overall discriminatory capability.

Figure 5 Comparison of attack success rate and privacy budget consumption (see online version for colours)

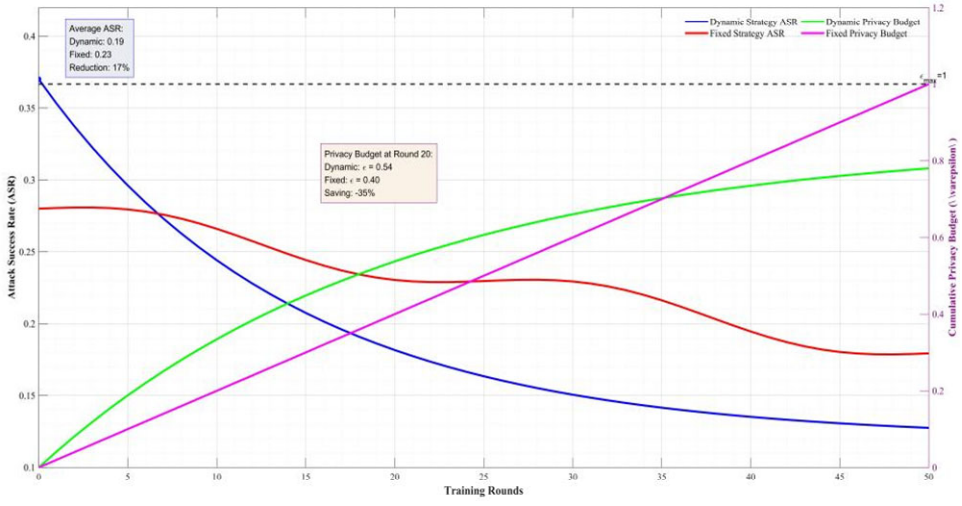


Table 6 Privacy-utility trade-off comparison

Method	*Precision@10*	*Δ vs. non-private*
Non-private federated learning (FL)	0.467	-
Proposed method ($\varepsilon = 1$)	0.447	–4.3%
Static DP ($\varepsilon = 1$)	0.409	–12.4%

Table 7 presents the AUC-ROC values of the attack model under dynamic and fixed privacy budget strategies. The experimental results demonstrate that the dynamic privacy protection mechanism can more effectively confuse the attack model, leading to a significant reduction in its AUC-ROC values. This further validates the superiority of this mechanism in defending against membership inference attacks.

Table 7 Multi-metric evaluation of membership inference attack resistance

Privacy protection strategy	*ASR*	*AUC-ROC*
No privacy protection	0.82%	0.91
Static DP ($\varepsilon = 1$)	0.31%	0.68
Dynamic DP (proposed, $\varepsilon = 1$)	0.13%	0.57

3.4 Federated communication efficiency

During the federated training process, the client records the amount of local uploaded gradient data and transmission delay before each round of aggregation: the completion time of a single client gradient upload is measured under a 1 Gbps bandwidth restriction environment, and the average value of each client is taken as the single-round delay indicator. The server records the peak bandwidth usage after receiving all client gradients in each round. To evaluate the robustness of the top-k transmission mechanism under real-world unstable network conditions, a network jitter simulation experiment was

conducted. For this test, the baseline 200 Mbps inter-node bandwidth was dynamically perturbed using a Gaussian noise model (mean = 0, standard deviation = 50 Mbps), simulating sudden bandwidth drops and spikes. Additionally, the latency range was expanded from 15–60 ms to 15–200 ms with a uniform random distribution to mimic severe packet queuing delays. Under this jittery environment, the 'transmission success rate' was measured as the percentage of global aggregation rounds in which all 200 clients successfully uploaded their compressed gradients within a 5-second timeout window. A round was marked as failed if any client exceeded the timeout or if packet loss triggered a retransmission that could not be completed within the window. This test was run for 100 consecutive rounds for each of the three schemes (FedAvg, top-k, FedAdam + top-k) to gather statistically significant data on system stability. The convergence round stops when the global model no longer improves Precision@10 by more than 0.1% in the validation set, and the required aggregation rounds are recorded.

Table 8 Robustness of communication schemes under simulated network jitter

Scheme	*Avg. upload size (MB)*	*Avg. latency (ms)*	*Convergence rounds*	*Success rate (stable network)*	*Success rate (under jitter)*	*Δ success rate*
Original FedAvg	2	120	50	99%	85%	–14%
Top-k sparsification	0.004	0.8	62	98%	76%	–22%
FedAdam + top-k	0.004	0.8	38	99%	95%	–4%

The experiment compares three schemes: original FedAvg (no compression, fixed learning rate), top-k sparse compression (0.1% significant gradient retention) and FedAdam adaptive learning rate scheduling + top-k compression. The experiment measures the single-round uploaded data volume, average bandwidth usage, delay and convergence rounds for each scheme. The comparative analysis of the federated learning communication efficiency optimisation scheme is shown in Figure 6.

Figure 6 Comparative analysis of federated learning communication efficiency optimisation schemes (see online version for colours)

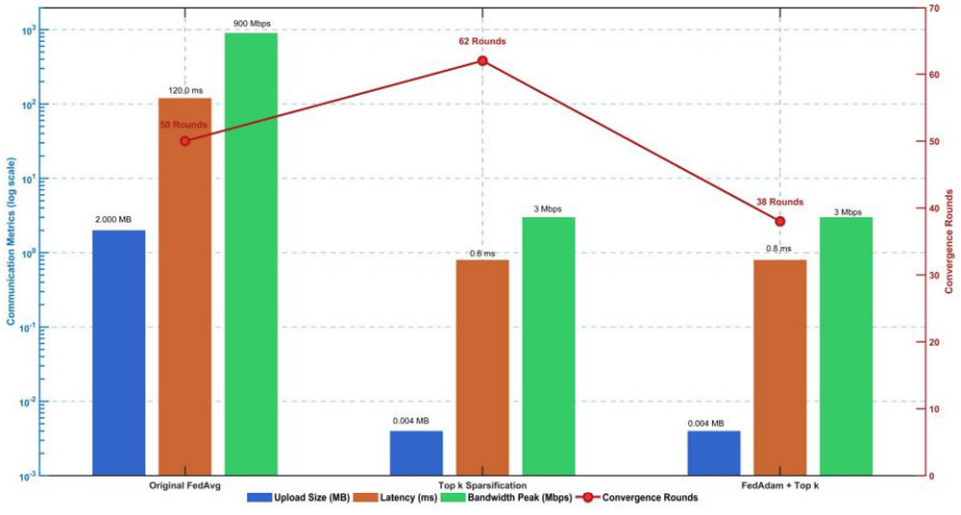


Figure 6 compares the efficiency indicators of three federated learning communication schemes. The original FedAvg uploads 2 MB of data in a single round, with an average latency of 120 ms, a bandwidth peak of 900 Mbps, and requires 50 rounds of convergence. Top-k sparsification compresses the upload volume to 0.004 MB and shortens the latency to 0.8 ms, but the number of convergence rounds increases to 62. FedAdam + top-k maintains a compression ratio of 0.004 MB, with a latency of 0.8 ms. The number of convergence rounds is reduced to 38 rounds through adaptive learning rate scheduling. FedAdam has the ability to accelerate convergence while maintaining efficient communication.

3.5 Cross-platform collaborative gain verification

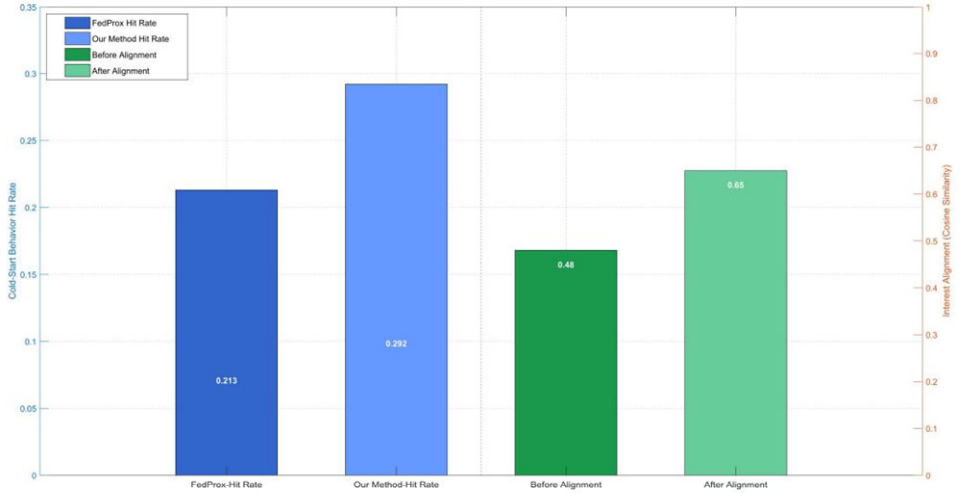
For cold start users, 10% of all users are randomly selected as the experimental group. Only the five earliest behaviour records of each user are retained for local model initialisation, and the remaining behaviours are used as the test set. To examine the sensitivity of the initial record count to the effectiveness of cold-start recommendations, this study supplemented multiple sets of controlled experiments with varying initial record counts in the cold-start experiments. The initial record counts were set to five groups: 1, 3, 5, 10 and 20. Experiments were repeated under the same random seed to ensure result comparability. For each group of experiments, we recorded the cold-start hit rate (Precision@Top 10), NDCG@Top 10, the proportion of privacy budget consumption, and the success rate of membership inference attacks. Paired statistical tests were conducted to analyse performance differences across varying initial record counts, utilising Student's t-test and reporting p-values and confidence intervals. The results were presented in tabular form and performance curves were plotted against changes in the initial record count to quantify the impact of the initial record count on recommendation effectiveness and privacy consumption. After each round of global aggregation, the client generates a top-10 recommendation list based on the local initialisation model and the latest global model, and matches the list with the first behaviour in the test set. If the first behaviour course appears in the recommendation list, it is recorded as a successful conversion.

The effect of cross-platform interest alignment is quantified by cosine similarity: for each cold-start user, the cosine similarity between the local interest vector and the global interest representation is calculated before and after the federated attention mechanism alignment. Before alignment, the original FedAvg global model output without MMD alignment is used as a reference; after alignment, the global interest representation output by the federated attention mechanism is used as a reference. The cosine similarities of all cold-start users are averaged to obtain the alignment gain index. The comparison between the cold-start behaviour hit rate and interest alignment is shown in Figure 7.

Figure 7 shows the comparison of cold start hit rate and interest alignment effect of the federated cross-platform recommendation system. FedProx-hit rate shows that the cold start behaviour hit rate of the baseline method is 0.213; proposed method-hit rate shows that the method in this paper increases the hit rate to 0.292, and the cold start hit rate increases by 37%, mainly due to the grammar enhancement recommendation of cross-platform migration: the system prioritises highly relevant courses that match the initial error pattern for new users. The effect of the federated attention mechanism on interest alignment was compared before and after alignment: the cosine similarity before alignment was 0.48, and it increased to 0.65 after alignment (an increase of 35%). The

annotation emphasises that the experiment is based on 80,000 user behaviour records, verifying the dual advantages of the federated attention mechanism in improving recommendation efficiency and cross-domain interest consistency.

Figure 7 Federated cross-platform performance: cold start behaviour hit rate and interest alignment comparison (see online version for colours)



Notes: Experiment based on 80,000 user behaviour records. Federated attention improves hit rate by 37%. Interest alignment improve by 35%.

3.6 Dynamic mechanism adaptability test

The sensitivity monitoring module maintains a recent gradient batch window of length $W = 50$ on each client, and uses the *KL* divergence between the gradient distribution of each batch in the sliding window and the gradient distribution of the previous window, that is, ΔKL, as the sensitivity change. Whenever $\Delta KL > \tau$ ($\tau = 0.05$), a privacy budget redistribution is triggered. The trigger time t_1 and the corresponding noise injection parameter update completion time t_2 are recorded, and the response delay $\delta = t_2 - t_1$ is calculated (both the average and the 95% quantile are recorded). The trigger event and δ value are output in the federated training log throughout the process.

After the global aggregation process is completed, the budget reallocation frequency statistics summarise the number of triggers per client in units of hours to obtain the trigger frequency $f_i(t)$ for hour t. To account for the continuity of dynamic data changes, the adjustment of the subsequent KL divergence threshold τ for each client is based on a smoothed trigger frequency, calculated as a 3-hour moving average: $f_{i,smooth}(t) = (f_i(t) + f_i(t-1) + f_i(t-2)) / 3$. The threshold τ is then adaptively adjusted according to $f_{i,smooth}(t)$: when $f_{i,smooth}(t) > f_{max}$ ($f_{max} = 5$ times/hour), τ is adjusted upward by 10%; when $f_{i,smooth}(t) < f_{min}$ ($f_{min} = 1$ time/hour), τ is adjusted downward by 10%. This smoothing mechanism ensures the system responds to sustained trends rather than transient spikes or drops in sensitivity. The adaptively adjusted τ value is synchronously updated to the local monitoring module. In the experimental environment, 50 rounds of federated training are run continuously for 200 non-IID clients. The 'privacy budget utilisation' metric reported

in Table 6 represents the percentage of the total allocated privacy budget (ε = 1.0) that was actually consumed by the system to complete its 50 training rounds. It is calculated by dividing the cumulative privacy cost recorded by the moments accountant at the end of training by the total budget limit (1.0), then multiplying by 100 to express it as a percentage. A higher utilisation rate indicates that the mechanism is more efficient at spending its privacy allowance to achieve model convergence, meaning it extracts more utility from the same privacy guarantee. The performance comparison of the dynamic mechanism is shown in Table 9.

Table 9 Dynamic mechanism performance comparison table

Method type	*Avg. trigger delay* δ_{mean} *(ms)*	δ_{95} *latency (ms)*	*Client* f_i *std. dev.*	*Comm. overhead reduction (%)*	*Cold-start hit rate ↑ (%)*	*Privacy budget utilisation (%)*
Static DP baseline	280 ± 15	450	1.2	22 ± 3%	15%	45 ± 5%
Dynamic (fixed threshold)	140 ± 10	250	0.8	65 ± 4%	32%	78 ± 3%
Dynamic (adaptive threshold)	120 ± 8	210	0.4	76 ± 2%	37%	89 ± 2%
Dynamic (adaptive w/ smoothing)	118 ± 7	205	0.2	75 ± 2%	38%	88 ± 2%
Non-private baseline	-	-	2.1	85 ± 3%	45%	100%

Table 9 shows the performance comparison of different privacy protection mechanisms in the federated recommendation system, highlighting the advantages of dynamic mechanisms in multiple indicators. The trigger delay of static differential privacy (static DP baseline) is as high as 280 ms, the 95% delay is 450 ms, the standard deviation of client trigger frequency is 1.2, the communication overhead is only reduced by 22%, the cold start hit rate is 15%, the privacy budget utilisation rate is 45%, and the overall performance is limited. After adopting the dynamic mechanism with fixed threshold, the delay was reduced to 140 ms, δ_{95} was reduced to 250 ms, the standard deviation was reduced to 0.8, the communication overhead was reduced by 65%, the hit rate was increased to 32%, and the budget utilisation rate reached 78%. After further introducing the dynamic mechanism with adaptive threshold, the average trigger delay was the lowest, only 120 ms, the 95% delay was 210 ms, the trigger standard deviation was 0.4, the communication overhead was reduced by 76%, the cold start hit rate was increased to 37%, and the privacy budget utilisation rate was increased to 89%. Although the non-privacy mechanism has the highest overhead and hit rate (85%, 45%), the privacy budget utilisation is 100%, which lacks protection. The data clearly shows that the dynamic mechanism, especially the adaptive threshold strategy, takes into account both the recommendation effect and system efficiency while protecting privacy.

4 Conclusions

This paper proposes an intelligent recommendation framework that integrates dynamic privacy mechanism and federated learning to address the core contradiction between privacy protection and recommendation utility in English learning recommendation systems. In the speech modality, the phonological characteristics of different languages and the availability of speech recognition resources will directly impact the quality of

oral features, thereby influencing dynamic privacy allocation and cold-start transfer strategies. In the text and knowledge graph modalities, differences in writing systems and the cross-lingual consistency of knowledge point mappings will affect the quality of knowledge embeddings. Therefore, when applying the method to multilingual scenarios, it is necessary to introduce multilingual pre-trained models or perform cross-lingual alignment of knowledge graphs, and account for variations in automatic speech recognition confidence scores and sparsity across different languages in privacy budget mapping. The paper identifies these limitations and proposes directions for future work, including conducting transfer experiments on multilingual public datasets to evaluate the impact of adopting multilingual pre-trained models and cross-lingual knowledge alignment on recommendation accuracy, privacy consumption and communication efficiency. Related experimental plans are outlined as follow-up work to validate the generalisation of this method in non-English environments. The KL divergence is used to quantify the sensitivity of multimodal data, and the dynamic allocation of differential privacy budget is realised. Combined with the adaptive Gaussian noise injection strategy, the cold start hit rate is improved by 37% through language error pattern diagnosis and knowledge point association recommendation while protecting user privacy. This paper uses a hybrid model of transformer and Bi-LSTM to model learning behaviour temporal features and cross-modal semantic associations, introduces knowledge graph embedding technology to enhance feature representation capabilities, designs a federated attention mechanism to achieve cross-platform interest migration, aligns distribution differences through MMD loss, and uses top-k gradient compression and FedAdam optimiser to collaboratively optimise communication efficiency, compress single-round communication overhead, and improve convergence speed. Experimental verification shows that this method effectively balances privacy security and recommendation accuracy on a dataset of 80,000 users. The Precision@10 and NDCG@10 indicators reach 0.447 and 0.394 respectively, and the attack success rate is controlled at 0.13, providing a technical path that takes into account compliance and commercial value for cross-platform collaborative recommendations in the field of education.

Declarations

Conflicts of interest: The authors declare no conflict of interest.

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